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EXAM : END TERM EXAM

Subset 11 – Web Frontend with AI

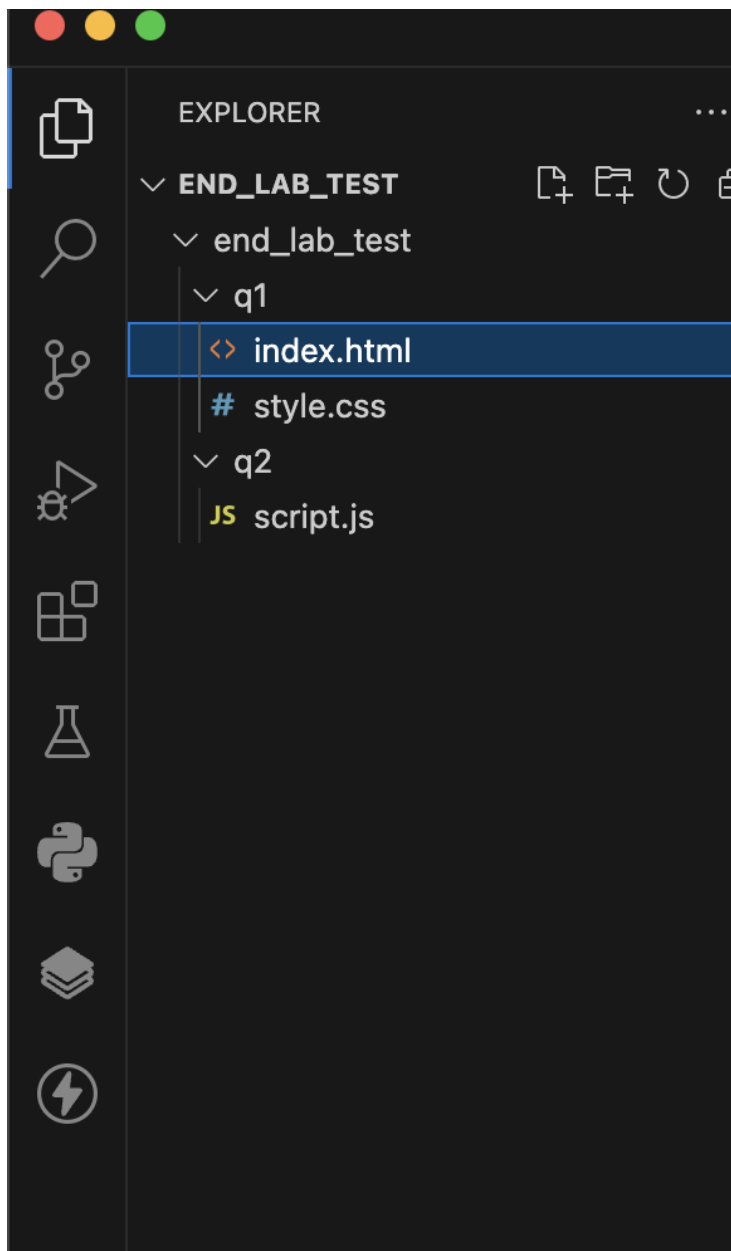
Q1: Create responsive student portal

- Task 1: Use AI to generate HTML structure.
- Task 2: Generate CSS grid layout.

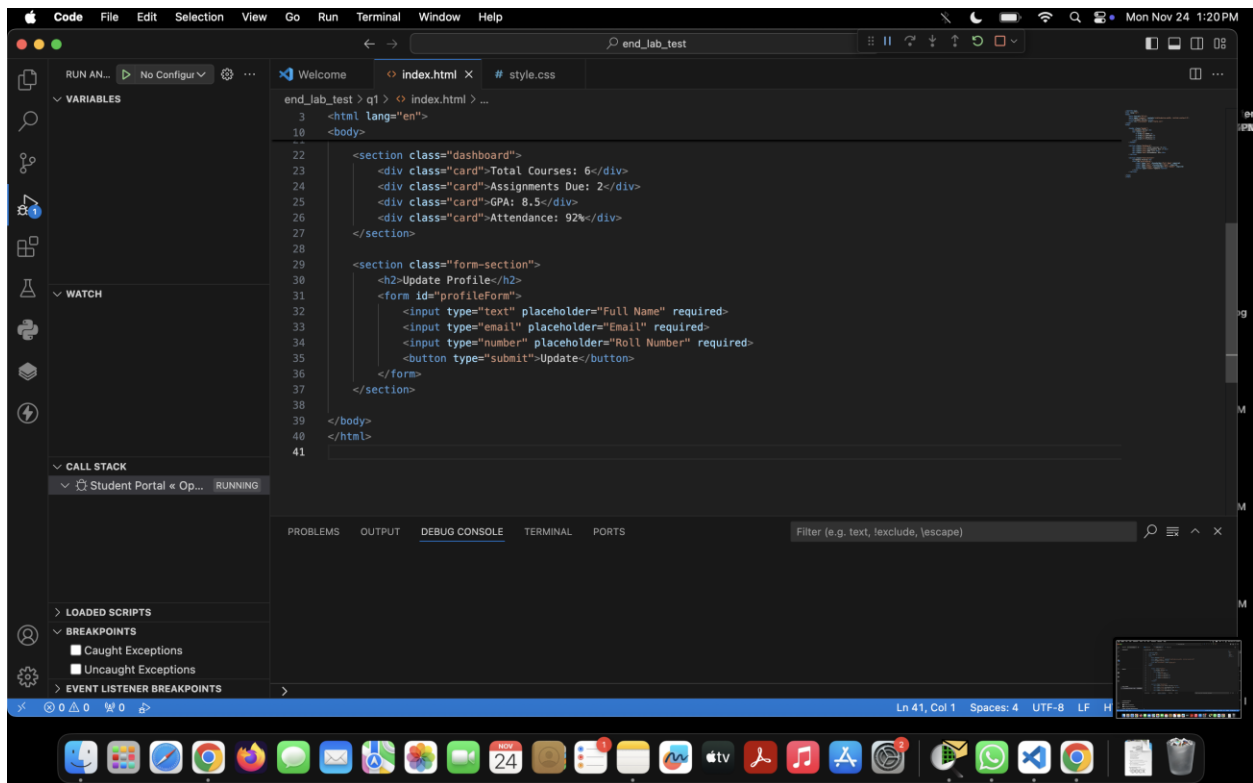
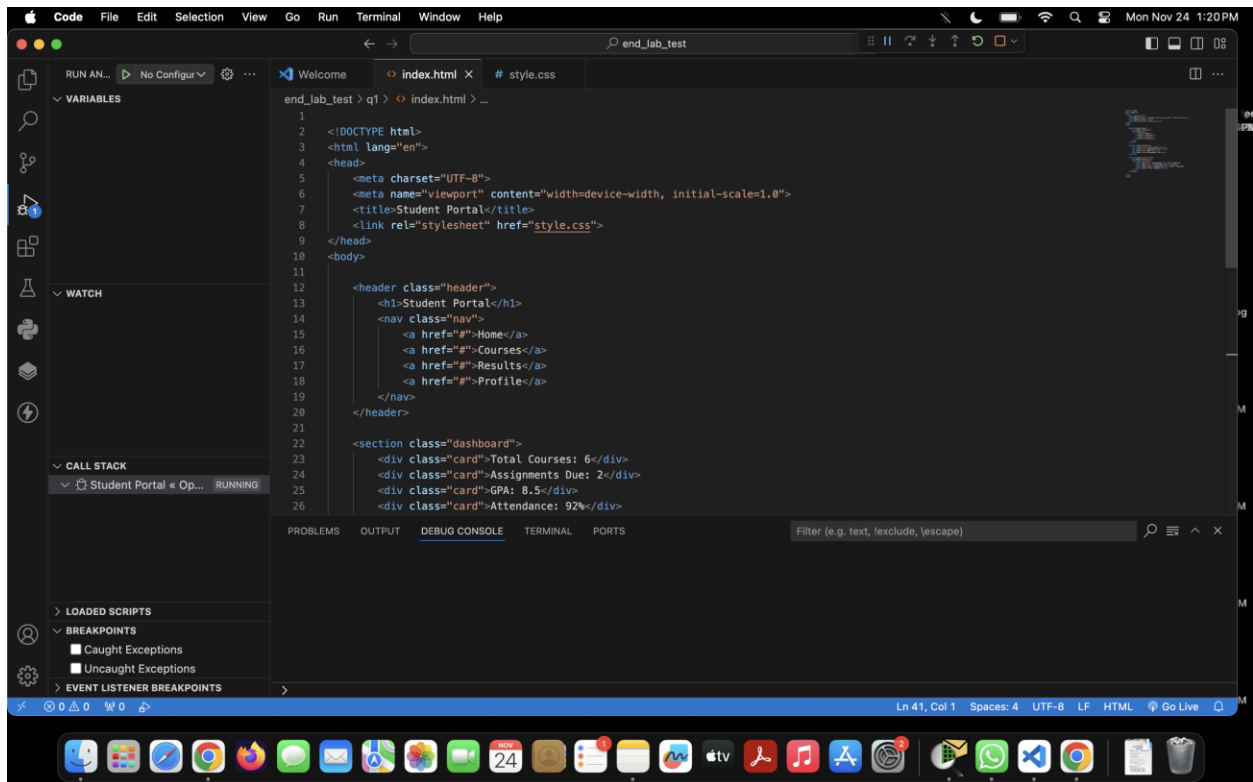
CODE PROMPT :

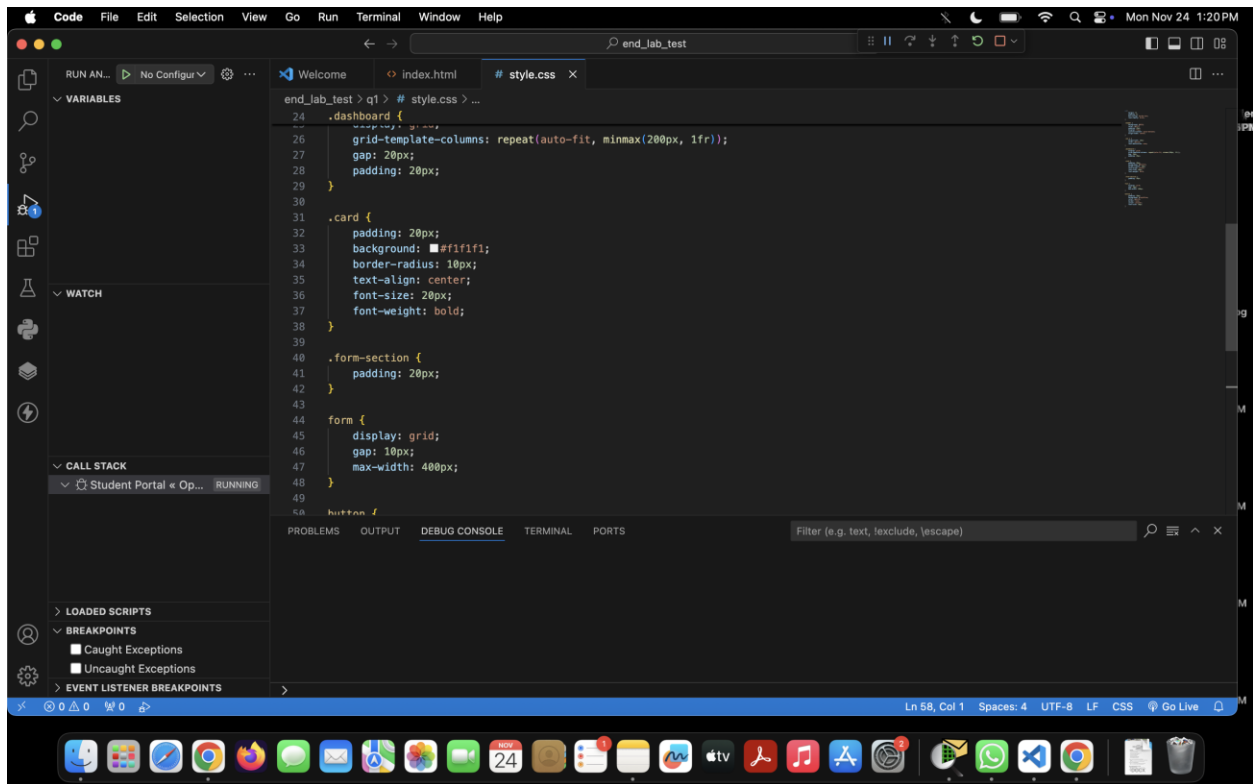
To Use AI to generate a responsive HTML structure for a Student Portal with a header, navigation menu, dashboard cards, and a profile update form in a simple way and generate a proper webpage

FOLDER STRUCTURE :

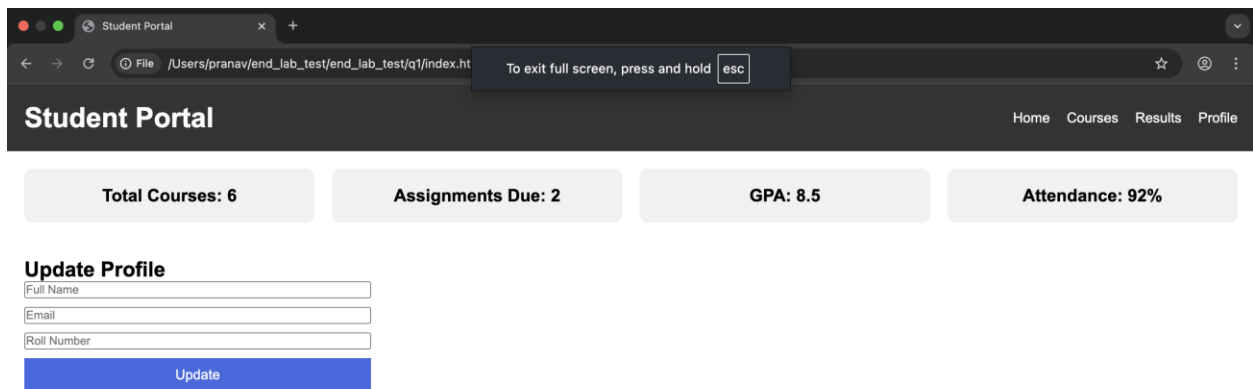


CODE GENERATED :





OUTPUT GENERATED :



OBSERVATION :

In this task, AI was used to generate a responsive Student Portal webpage.

AI created the complete HTML structure including header, navigation, cards, and a form.

The HTML used semantic tags for better readability and organization.

AI also generated CSS using a Grid layout for modern responsiveness.

The dashboard cards automatically adjust to screen size using auto-fit and minmax().

The layout became mobile-friendly without writing manual breakpoints.

The form section was styled with clean spacing and structured inputs.

The overall design looked clean, balanced, and professional.

AI helped complete the frontend quickly while maintaining quality.

This shows how AI can speed up web development and improve layout designs.

Q2: Add interactive JS

- Task 1: Use AI to write form validation logic.
- Task 2: Add event listeners for UI elements.

CODE PROMPT :

Write JavaScript to validate the profile update form and make the page interactive.

The script should check if the fields are filled correctly and show alerts if something is wrong.

Also add event listeners so the form responds when the user types or clicks the update button.

CODE GENERATED :

The screenshot shows the VS Code editor with the file explorer on the left. The project structure includes `END_LAB_TEST`, `.vscode`, `launch.json`, `end_lab_test` (containing `q1` and `q2`), `index.html`, `style.css`, and `JS script.js`. The `JS script.js` file is open in the editor, showing the following code:

```
1
2
3 function validateForm() {
4     let name = document.getElementById("name").value.trim();
5     let email = document.getElementById("email").value.trim();
6     let roll = document.getElementById("roll").value.trim();
7
8     if (name === "" || email === "" || roll === "") {
9         alert("All fields must be filled!");
10        return false;
11    }
12
13    let emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
14    if (!emailPattern.test(email)) {
15        alert("Please enter a valid email address!");
16        return false;
17    }
18
19    if (isNaN(roll)) {
20        alert("Roll number must contain numbers only!");
21        return false;
22    }
23
24    alert("Profile Updated Successfully!");
25    return true;
26 }
27
28
29
30 document.getElementById("updateBtn").addEventListener("click", function (e) {
31     e.preventDefault();
32     validateForm();
33 })
```

The terminal at the bottom shows the following commands and output:

```
pranav@PRANAVs-MacBook-Air end_lab_test % cd ~END_LAB_TEST/q1
zsh: no such user or named directory: END_LAB_TEST
pranav@PRANAVs-MacBook-Air end_lab_test % cd ~/end_lab_test/q1
cd: no such file or directory: /Users/pranav/end_lab_test/q1
pranav@PRANAVs-MacBook-Air end_lab_test % cd ~/end_lab_test/end_lab_test/q1
pranav@PRANAVs-MacBook-Air q1 % open index.html
pranav@PRANAVs-MacBook-Air q1 % open index.html
pranav@PRANAVs-MacBook-Air q1 %
```

The screenshot shows the VS Code editor with the file explorer on the left. The project structure is the same as the first screenshot. The `JS script.js` file is open in the editor, showing the following code:

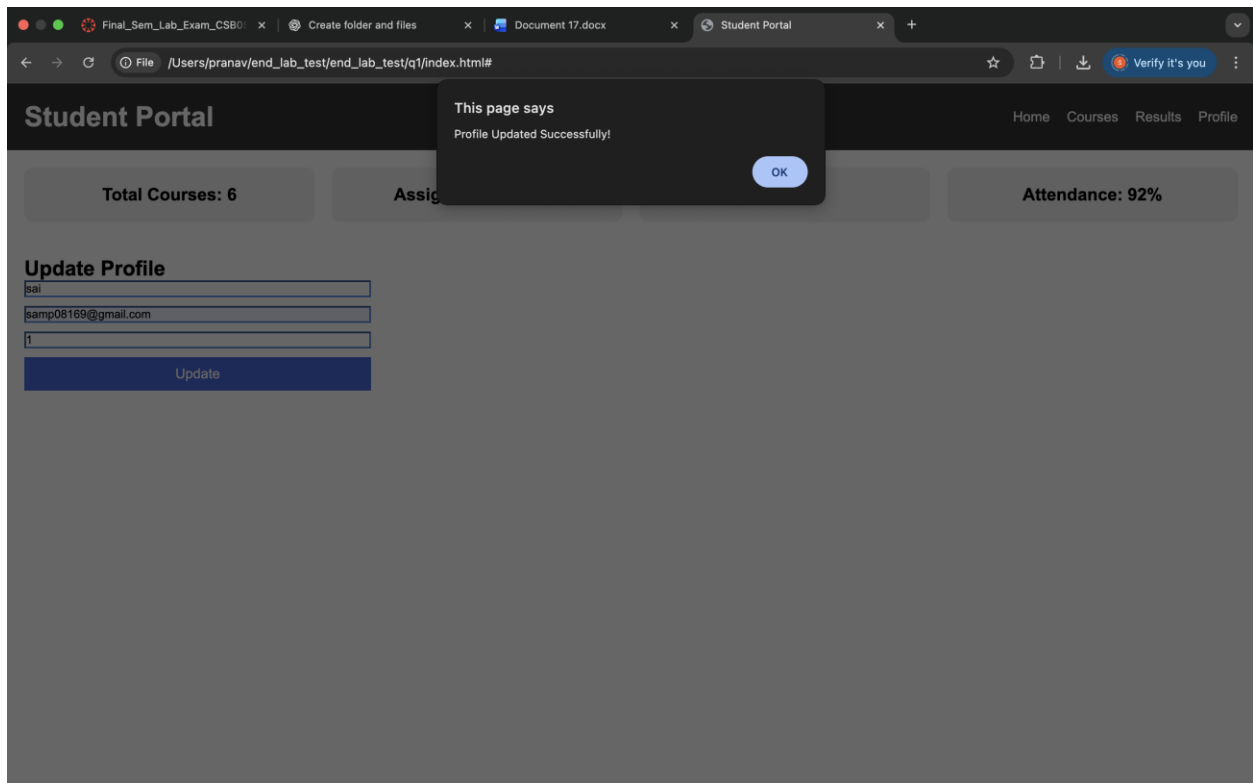
```
24     alert("Profile Updated Successfully!");
25     return true;
26 }
27
28
29
30 document.getElementById("updateBtn").addEventListener("click", function (e) {
31     e.preventDefault();
32     validateForm();
33 });
34
35 document.querySelectorAll("input").forEach(input => {
36     input.addEventListener("input", function () {
37         this.style.borderColor = "#4A90E2";
38     });
39 });
40
```

The terminal at the bottom shows the following commands and output:

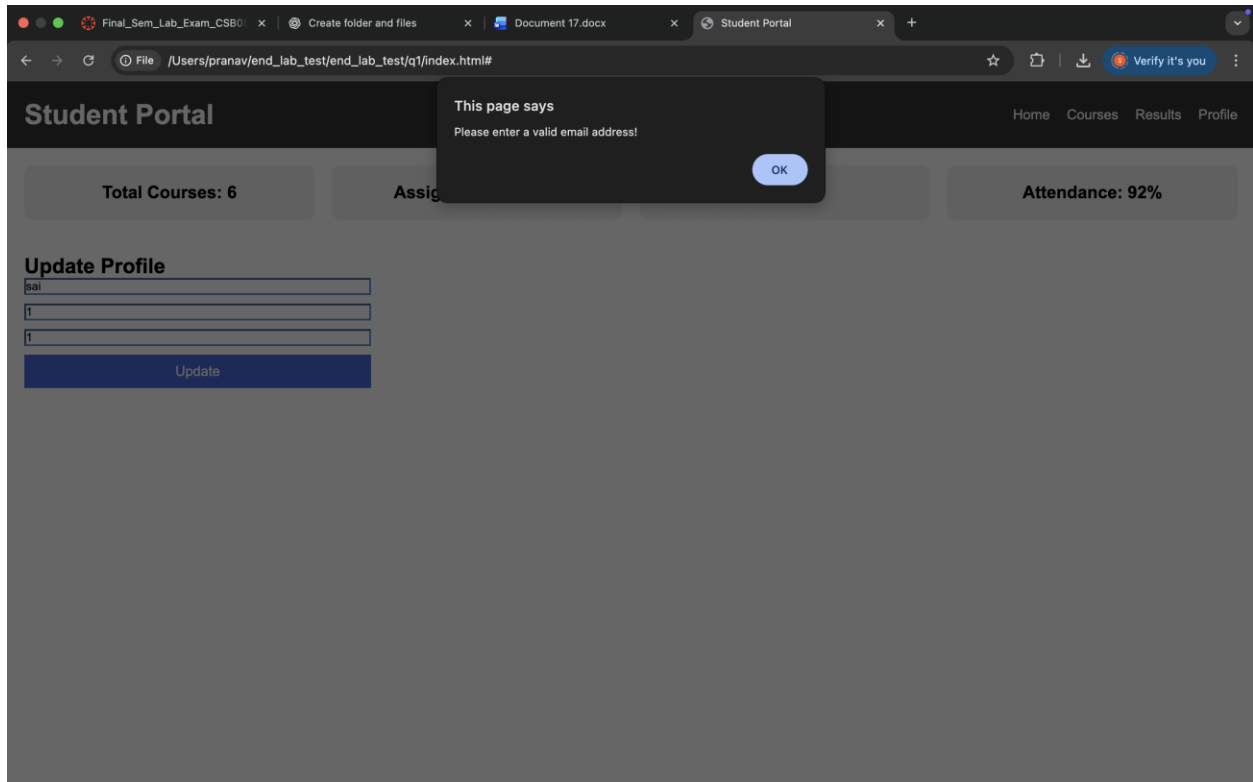
```
pranav@PRANAVs-MacBook-Air end_lab_test % cd ~END_LAB_TEST/q1
zsh: no such user or named directory: END_LAB_TEST
pranav@PRANAVs-MacBook-Air end_lab_test % cd ~/end_lab_test/q1
cd: no such file or directory: /Users/pranav/end_lab_test/q1
pranav@PRANAVs-MacBook-Air end_lab_test % cd ~/end_lab_test/end_lab_test/q1
pranav@PRANAVs-MacBook-Air q1 % open index.html
pranav@PRANAVs-MacBook-Air q1 % open index.html
pranav@PRANAVs-MacBook-Air q1 %
```

OUTPUT :

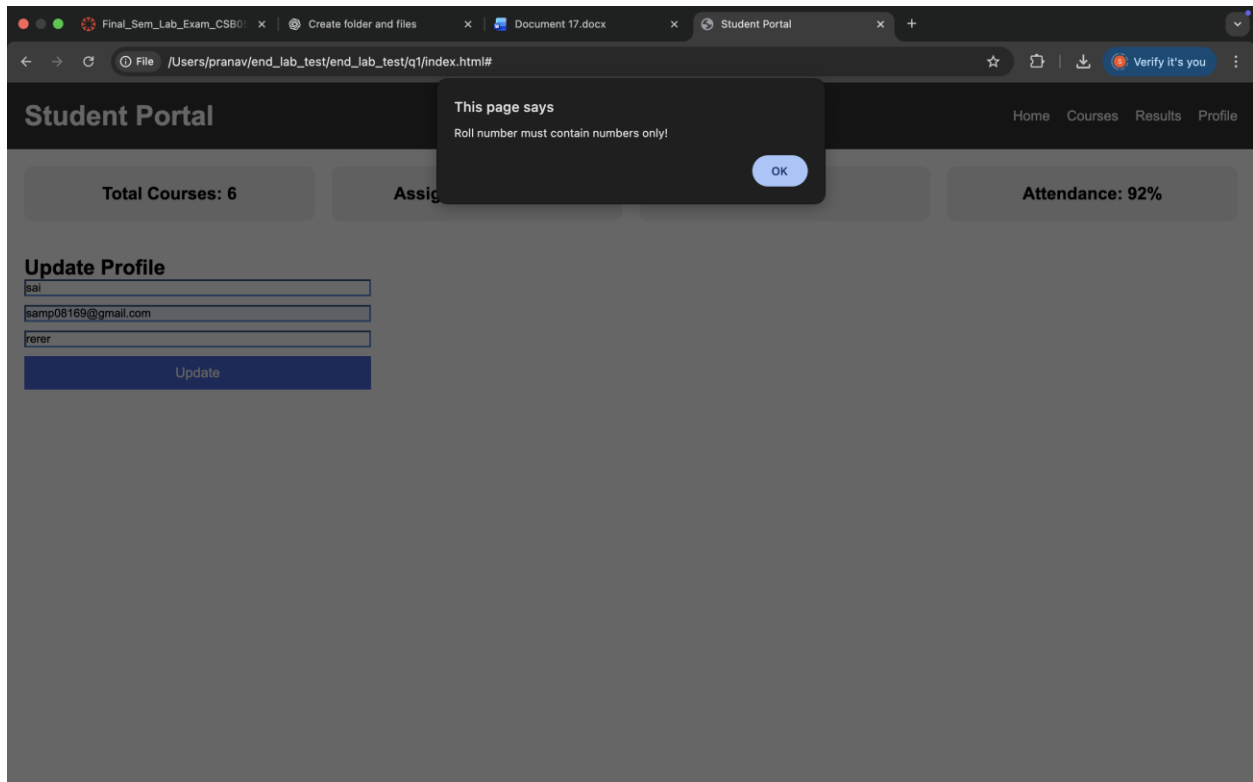
UPDATED SUCCESSFULLY VALIDATION



TO ENTER A VALID EMAIL ADDRESS:



VALIDATE A ROLL NUMBER :



OBSERVATION:

In this task, JavaScript was used to add interaction to the student portal.

The script validates the form by checking empty fields, email format, and roll number.

If the input is wrong, the user gets an alert message.

Event listeners were added so the button click triggers validation.

Input fields also highlight while typing, improving the user experience.

The output shows that the validation and interactions work correctly.